Experiment 09

Design a simple topology and configure with one router, two switches and PCs using Cisco Packet Tracer

Aim

To construct two switched LANs interconnected by one router and verify communication in Packet Tracer.

Requirements

Packet Tracer; one 2911 router, two 2960 switches and four PCs.

Procedure / algorithm

1. Place devices and cable them using topology.txt.
2. Assign each PC its unique IPv4 address, mask and router-interface gateway from addressing.csv.
3. Apply R1.cfg and switch.cfg; ensure both routed interfaces are administratively enabled.
4. Wait for links to turn green; verify connected routes and MAC learning.
5. Ping within each LAN and across LANs; inspect ARP/ICMP in Simulation mode.

Complete source / configuration: addressing.csv

device	interface	IPv4	mask	gateway
R1	GigabitEthernet0/0	192.168.10.1	255.255.255.0	none
R1	GigabitEthernet0/1	192.168.20.1	255.255.255.0	none
PC1	FastEthernet0	192.168.10.11	255.255.255.0	192.168.10.1
PC2	FastEthernet0	192.168.10.12	255.255.255.0	192.168.10.1
PC3	FastEthernet0	192.168.20.11	255.255.255.0	192.168.20.1
PC4	FastEthernet0	192.168.20.12	255.255.255.0	192.168.20.1

Complete source / configuration: R1.cfg

```
enable
configure terminal
hostname R1
interface GigabitEthernet0/0
 ip address 192.168.10.1 255.255.255.0
 no shutdown
exit
interface GigabitEthernet0/1
 ip address 192.168.20.1 255.255.255.0
 no shutdown
end
copy running-config startup-config
show ip interface brief
show ip route
```

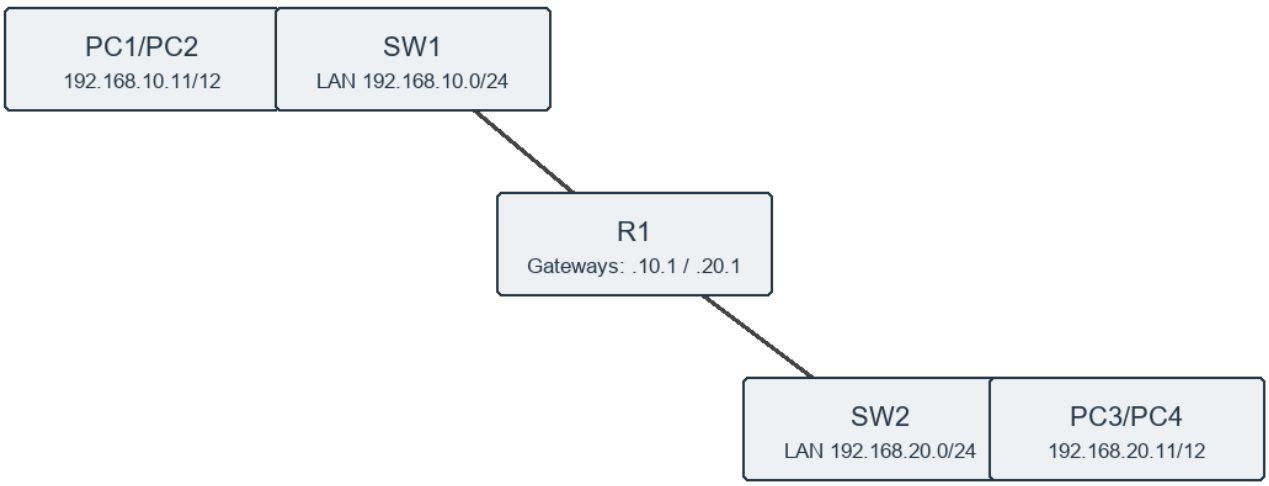
Complete source / configuration: switch.cfg

```
enable
configure terminal
interface range FastEthernet0/1-3
 switchport mode access
 no shutdown
end
copy running-config startup-config
show mac address-table
```

Complete source / configuration: topology.txt

```
PC1 --- Fa0/1 SW1 Fa0/3 --- Gi0/0 R1 Gi0/1 --- Fa0/3 SW2 Fa0/1 --- PC3
PC2 --- Fa0/2 SW1                                SW2 Fa0/2 --- PC4
All links use copper straight-through Ethernet cables.
R1: 2911; SW1/SW2: 2960. Apply switch.cfg to both switches.
PC1: ping 192.168.10.12
PC1: ping 192.168.20.11
PC4: ping 192.168.10.11
```

Diagram: topology



Lab design / synthetic fixture; not a simulator execution screenshot

Actual output / verification

Not executed in the required lab/tool environment. No actual operational output is claimed.

Expected output - not an execution record

Both connected networks appear in the routing table; after ARP resolution all PCs can communicate.

Observations

No static routes are needed because both subnets are directly connected to R1. Hosts need their local router address as the gateway.

Result

The complete topology and device configuration are prepared. Packet Tracer execution remains pending.